Nathan Owen Santoso

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PROFESSIONAL SUMMARY

Full-stack software engineer with strong foundations in Java, JavaScript/React, and SQL, experienced delivering production changes in complex, long-lived systems. Known for careful debugging, requirements analysis, and clear cross-functional communication. JLPT N2 and native English speaker with 2 years of total professional work experience.

EXPERIENCE

LogicVein — **Mar 2025 – Oct 2025**

Software Engineer — *Tokyo, Japan*

- Developed and maintained full-stack features across Java/Kotlin backend and JavaScript/React frontend in a 10+ year legacy codebase
- Integrated REST API endpoints with UI interactions, ensuring correct data flow, state handling, and reliable user-facing behavior
- Diagnosed and resolved UI defects and race conditions by reproducing issues, tracing application behavior, and implementing reliable fixes
- Collaborated in Japanese with QA to reproduce issues, validate fixes, and iterate on feature delivery through feedback cycles

Jacobs Engineering — **Nov 2022 – Feb 2024**

Electrical Engineering Intern | Building Services — *Australia*

- Led data-driven analysis for solar panel consultation evaluation, resulting in 15% cost savings and 20% energy savings
- Developed detailed electrical schedules for 50+ LV circuits, improving project clarity and reducing coordination errors

UWA Young Engineers (UWAYE) — **Jun 2019 – Jun 2023**

Project Lead & Digital Team | Professional Development — *Australia*

- Led a team of 6 to host an EV case study with ASX-listed company, with a turnout of 80+ students and company representatives
- Organized and executed professional development events for 300+ STEM students, enhancing industry networking opportunities and career prospects

Exida — **Nov 2021 – Feb 2022**

Internship | Engineering Risk — *Australia*

- Streamlined safety certificate creation process through Python automation
- Supported development of electrical safety training for Rio Tinto, following IEC 61508 and IEC 61511 standards
- Enhanced safety training material by redesigning PowerPoint slide graphics

PROJECTS

UWA Research Funding Opportunities Website Rebuild

MySQL, AWS, Python, Django | Database Design, AWS EC2 Deployment

- Led the design and implementation of a secure SQL relational database, ensuring secure storage of user data and funding opportunity listings
- Deployed the application to an AWS EC2 instance and implemented Microsoft OAuth authentication, enhancing security and accessibility for users
- Contributed to the Python Django backend and integration with the React JS front-end
- Conducted requirements analysis and liaised directly with the client to translate needs into implementation

OpenAI-Powered Video Automated Caption and Transcription

Powershell, Python | OpenAI Whisper, Signal processing

- Implemented OpenAI's audio-to-text model Whisper for lecture videos, adding automated word-level highlighted captions and full lecture transcriptions
- Utilised noise reduction and frequency boosting of vocal frequencies to further increase intelligibility, allowing for greater acquisition of educational material and assisting hearing impaired students

Battery Management System

PCB Design, Risk Management, Integrated Circuits

- Designed and implemented SPI-based communication for BQ76942 Battery Monitor IC integration with microcontroller, optimizing data exchange efficiency
- Created and maintained a comprehensive Risk Register, ensuring project milestones met risk management standards
- Conducted requirements analysis, technical documentation, and communication protocol configurations, leveraging Texas Instruments resources and technical manuals

Audio Super-Resolution using Time and Frequency Generative Adversarial Networks

Python, Tensorflow | AI, Signal Processing, Neural Networks

- Combined state-of-the-art AI GAN models with Time and Frequency domain approaches for audio super-resolution
- Implemented Tensorflow deep learning methodologies for audio processing and neural network hyperparameter optimization
- Synthesised two state-of-the-art audio super-resolution techniques (time and frequency domain, and using GANs) to produce a novel technique

Autonomous Robotics Boat

C++, Arduino | Circuit design, Embedded systems

- Repurposed RC boat to autonomously drive forward and return to set position using closed-loop bang-bang control system
- Designed, soldered, and implemented circuit board design
- Programmed Arduino (C++) Atmega328P microcontroller to utilize sonar sensor, magnetosensor, rotor

EDUCATION

The University of Western Australia **2018 – 2023**

Master of Professional Engineering | Electrical and Electronic Engineering *Australia*

- 6.7/7.0 GPA | 82.9 WAM
- UWA Engineering Scholarship
- Competed in WA Volleyball League as a member of the UWA Volleyball Club

Bachelor of Philosophy (First Class Hons) | Computer Science and Engineering *Australia/Japan*

- 6.9/7.0 GPA | 86.7 WAM
- Studied abroad in Nishinomiya, Japan at Kwansei Gakuin University
- Research thesis explored implementing deep learning AI for audio super-resolution

SKILLS, QUALIFICATIONS & INTERESTS

- **Languages/Frameworks: Java; React; JavaScript;** Python; C; SQL; Matlab; Arduino
- **Technical:** MS Excel; Cadence; Data Visualization; Machine Learning
- **Qualifications: JLPT N2**; AMEB Piano Grade 7, Engineers Australia
- **Interests:** Language learning; Karaoke, Piano; Badminton; Consumer Technology, AI